

Effects of substrate availability and competing vegetation on natural regeneration of white spruce on logged boreal mixedwood sites

Nicola A. Kokkonen, S. Ellen Macdonald, Ian Curran, Simon M. Landhäusser, and Victor J. Lieffers

Abstract: Given a seed source, the quality of available substrates is a key factor in determining the success of white spruce (*Picea glauca* (Moench) Voss) natural regeneration. We examined the influence of substrate and competing vegetation on survival and growth of natural regeneration of white spruce up to 4 years following harvesting in deciduous-dominated upland boreal mixedwood sites. Feather moss, thick soil surface organic layers, litter, and solid wood were poor substrates for establishment. Early successional mosses establishing on mineral soil, thin organics, and rotten wood were generally favourable microsites but were not highly available on postharvest sites. Mineral soil substrates were not as suitable as expected, likely because on a postlogged site, they are associated with unfavourable environmental characteristics (e.g., low nutrient availability, exposure). There was some evidence that survival and growth of seedlings were improved by surrounding vegetation in the first years, but heavy competing vegetation had a negative impact on older seedlings. Burial by aspen litter greatly increased seedling mortality, especially when combined with a brief period of submergence due to heavy spring snowmelt. The results provide insight into conditions under which natural regeneration could be an option for establishing white spruce following harvesting of deciduous-dominated boreal mixedwood forests.

Key words: natural regeneration, white spruce, substrate, seedling survival, competing vegetation, forest management.

Résumé : Pour une source de graines donnée, la qualité des substrats disponibles est un facteur clé pour le succès de la régénération naturelle d'épinette blanche (*Picea glauca* (Moench) Voss). Nous avons étudié l'influence du substrat et de la végétation concurrente sur la survie et la croissance de la régénération naturelle d'épinette blanche jusqu'à 4 ans après la coupe de peuplements mixtes boréaux dominés par des espèces décidues et établis sur des stations bien drainées. Les mousses hypnaces, les couches épaisses de matière organique sur le sol, la litière et le bois massif constituaient de mauvais substrats pour l'établissement des semis. Les mousses de début de succession établies sur le sol minéral, les couches minces de matière organique sur le sol et le bois décomposé constituaient généralement des microsites favorables, mais étaient peu disponibles sur les stations récoltées. Les substrats formés de sol minéral n'étaient pas aussi appropriés que prévu, probablement parce que sur une station qui a été récoltée, ils sont associés à des caractéristiques environnementales défavorables (p. ex. : faible disponibilité des nutriments, exposition). La survie et la croissance des semis ont montré des signes d'amélioration liés à la présence de végétation environnante au cours des premières années, mais une forte compétition exercée par cette végétation a eu un impact négatif sur les plus vieux semis. L'ensevelissement des semis par la litière de peuplier a considérablement augmenté leur mortalité, particulièrement lorsqu'il était jumelé à une brève période d'immersion causée par une importante fonte des neiges au printemps. Les résultats donnent un aperçu des conditions dans lesquelles la régénération naturelle pourrait constituer une option pour l'établissement de l'épinette blanche après la récolte des forêts mixtes boréales dominées par des espèces décidues. [Traduit par la Rédaction]

Mots-clés : régénération naturelle, épinette blanche, substrat, survie des semis, végétation concurrente, aménagement forestier.

Introduction

White spruce (*Picea glauca* (Moench) Voss) is an economically and ecologically important conifer that is widely distributed across the boreal forests of North America. On upland mixedwood sites that support healthy growth of both coniferous and broad-leaf species (MacDonald 1995), white spruce tends to function as a later successional species eventually gaining dominance over the

early successional canopy of deciduous (i.e., broadleaf) trees (Bergeron et al. 2014). White spruce dominated forests across Canada have mostly originated following fire (Kemball et al. 2006; Gärtner et al. 2011), and the coincidence of wildfire with mast seed years is key to successful natural regeneration of this species (Peters et al. 2005). While the factors affecting the early establishment of natural regeneration on unmanaged sites are well studied

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N.A. Kokkonen,* S.E. Macdonald,† I. Curran,† S.M. Landhäusser, and V.J. Lieffers. Department of Renewable Resources, University of Alberta, 751 General Services Building, Edmonton, AB T6G 2H1, Canada.

Corresponding author: S. Ellen Macdonald (email: ellen.macdonald@ualberta.ca).

*Present address: School of Forest Sciences, University of Eastern Finland, Yliopistokatu 7, Joensuu, 80101 Finland.

†Present address: P.O. Box 8535, STN Main, Canmore, AB T1W 2V3, Canada.

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(e.g., Gärtner et al. 2011 and citations therein), we know much less about natural regeneration of white spruce following harvesting. This is particularly true for boreal mixedwood sites in which broadleaf species (primarily trembling aspen (*Populus tremuloides* Michx.)) quickly dominate postharvest sites (but see Martin-DeMoor et al. 2010).

Suitable germination microsites and density of seed rain are likely the most important factors determining white spruce establishment in both intact forests (Nienstaedt and Zasada 1990; DeLong et al. 1997) and in recently burned stands (Greene et al. 1999, 2007; Charron and Greene 2002; Peters et al. 2005). Suitable substrates for white spruce germination and establishment in wildfire-origin stands include mineral soil, well-decomposed organics, and thin moss layers (Gärtner et al. 2011). Surrounding vegetation can strongly influence white spruce natural regeneration (Cole et al. 2003; Cater and Chapin 2000; Landhäusser et al. 1996), and environmental conditions such as flooding, frost, litter collection, and frost heaving reduce seedling establishment (Gärtner et al. 2011). The effect of surrounding vegetation on the growth and survival of seedlings is highly dependent on the nature of the surrounding species (Landhäusser et al. 1996) and the conditions that they create (Cater and Chapin 2000); nonlethal levels of surrounding vegetation tend to decrease seedling diameter growth (Groot 1999).

On cutover sites, harvesting leaves surface organics, including slash, feather mosses, and organic layers, more or less intact; thus little natural regeneration is expected on logged sites, especially on aspen-dominated sites, which have thick layers of leaf litter added each year (Greene et al. 1999; Wang and Kembell 2005). Natural regeneration of white spruce can be further inhibited by rapid establishment of competing vegetation, including aspen suckers and grasses such as *Calamagrostis canadensis* (Michx.) P. Beauv. (Lieffers et al. 1993). For these reasons, postharvest regeneration of white spruce on boreal mixedwood sites is primarily achieved through planting of seedlings, because natural regeneration is considered to be too unpredictable (Martin-DeMoor et al. 2010; Morimoto et al. 2017). However, as the cost of reforestation is high relative to the value of the wood harvested in boreal forests, natural regeneration is an economically attractive alternative (Gärtner et al. 2011). There is growing evidence that reliance on natural regeneration for white spruce on harvested sites could be a cost-effective strategy, in the right conditions. It has been found to regenerate on relatively undisturbed forest floor after logging of mixedwood stands (Wurtz and Zasada 2001), and even deciduous-dominated mixedwood stands can apparently support good regeneration of white spruce, despite relatively few seed-producing spruce trees nearby (Martin-DeMoor et al. 2010; Solarik et al. 2010). Overall, however, we have insufficient information about the success of, and factors influencing, germination and early establishment of naturally regenerated white spruce seedlings on the various substrates of cutover areas of boreal mixedwood forests.

Our objective was to determine how known influential factors — substrate and surrounding vegetation — impact the early success of naturally established white spruce seedlings. We used three approaches to fulfill this objective: a sowing study to assess the early establishment of seedlings as affected by substrate; surveys of natural regeneration following harvesting to assess the availability and suitability of different regeneration microsites; and vegetation surveys to quantify the effects of surrounding vegetation on growth and survival of seedlings. We hypothesized that decayed wood and thin organic substrates would result in higher germination and survival of white spruce but that these substrates would have low availability on logged sites. We expected grass to have more negative effects than forbs and that tall shrubs or aspen saplings could have a positive effect at lower densities but negative effects at high densities. These results can be used to determine the potential for use of natural regeneration as an

approach to reforestation in harvested deciduous-leading mixed-woods.

Methods

We investigated the establishment (germination and short-term survival) and early growth of white spruce seedlings after logging in four different regions — Drayton Valley, Edson, Grande Prairie, and Peace River — each with multiple sites. Part I of this section focusses on the early establishment of white spruce as a function of microsite from an experimental sowing study. Part II examines establishment of naturally regenerated white spruce as a function of substrate. Part III quantifies the subsequent success (survival and growth) of naturally regenerated seedlings as affected by competing vegetation.

All sites were classified as upland mixedwood sites (MacDonald 1995), and soils were generally relatively moist and rich. The study region in which the sites are located is, overall, dry (450–550 mm annual precipitation, of which ~70% is rain) with cool short summers (average May to August temperature 11–13 °C). All of the sites were dominated by trembling aspen prior to logging except at the Peace River location (56°N, 116°W), where experimental sowing was also done on sites that were formerly dominated by mature white spruce.

Part I: substrate effects on early establishment of sown seeds

We experimentally tested the influence of substrate on early establishment in both deciduous (aspen) and coniferous (white spruce) dominated sites in the Peace River region by means of an artificial sowing study begun in May of 2007. These sites had been clearcut harvested in the winter of 2005–2006. The spruce-dominated sites were site-prepared in the winter of 2006–2007 using a ripper plough, while the deciduous sites were left untreated. One transect was laid out at each of five coniferous cutblocks (cutblocks were ~25 ha, transects were each 450 m in length) and four deciduous cutblocks (cutblocks were ~10 ha, transects were each 45 m in length). On each transect, 10 equally spaced points were staked; eight of these were selected for seeding and two were left as unseeded controls to assess background natural regeneration rates. At each of the 10 points, we selected the nearest example of each of eight substrate types for the conifer sites and five substrate types for the deciduous sites (Table 1). Sixty seeds were sown in a 20 cm² area in each substrate type at each of the eight points per transect. The seeds originated from a local population (seed zone 71 in Alberta) and had been cold stratified prior to sowing. Seeds were sown by hand in early spring, and seeded areas were re-examined 4 months later for seedling establishment. There was almost no natural establishment of white spruce in the unseeded plots and thus we did not attempt to account for it; establishment was calculated as surviving seedlings as a percentage of seeds sown.

The experimental design was a randomized complete block in which each sample point (nested within transect) was treated as a block that contained one replicate of each substrate type. The effect of substrate on seedling establishment (percentage) was tested using an analysis of variance (ANOVA) for the deciduous- and conifer-dominated sites separately. Establishment (as a percentage of sown seeds) was the response variable, substrate type was the main (fixed) effect, and sample point within transect was included as a random (block) effect. Prior to these analyses, the data were log transformed to meet assumptions of normality and homoscedasticity of the model residuals. Post hoc comparisons among substrate types were conducted using lsmeans with a Bonferroni adjustment of alpha. An α level of 0.05 was chosen for all of our analyses. These analyses were conducted using PROC MIXED in SAS version 9.3 (SAS Institute, Inc. 2011).

Table 1. Substrate types on harvested deciduous- and conifer-leading boreal mixedwood sites.

Substrate	Description
Deciduous	
Rotten wood	Decaying wood that was dead and in the soft phase before harvest
Moss	Intact mosses >3 cm in depth from the preharvest forest (usually dying)
Thick organic (>5 cm)	Leaf litter layer over fine organic and humic layers with depth > 5 cm
Thin organic (<5 cm)	Leaf litter layer over a fine organic and humic layers with depth < 5 cm
Mineral soil	Mineral soil exposed through the disturbance of the LFH layer during harvesting
Solid wood	Coarse woody debris created by harvesting (no seeds sown on this substrate)
Coniferous	
Mound	Overtured raised layer of loose mineral soil displaced from trench and deposited next to scarification track
Rotten wood	Decaying wood that was dead before harvest and scraped clean of debris by scarification
Moss	Intact feather moss > 3 cm in depth from the preharvest forest (usually dying)
Litter	Needle litter layer over fine humic layers
Shelf	Flat area of broken up mineral soil higher than the scalp on the side of the trench
FH	Organic layer that was exposed after needle litter was scraped off
Scalped	Bladed exposed flat mineral soil
Trench	Cut depression with mineral soil exposed by site preparation
Slash	Small woody material and organic matter created during harvesting (no seeds sown on this substrate)
Solid wood	Coarse woody material created by harvesting (no seeds sown on this)

Note: For coniferous sites, moss (feather moss) and litter were naturally occurring substrates, while the others were the result of site preparation using a ripper plow.

Part II: substrate effects on establishment of natural seedlings

Natural white spruce seedling establishment was assessed by sampling along transects in harvested deciduous-dominated mixedwood stands (aspen – white spruce) selected in four regions of Alberta: Drayton Valley (53°N, 115°W), Edson (53°N, 116°W), Grande Prairie (54°N, 118°W), and Peace River (56°N, 116°W). The Drayton Valley and Edson region sites were harvested in the summer of 2003 prior to the release of a mast seed crop in fall 2003; therefore, the immediately dispersed seeds originated from the occasional residual white spruce trees still standing in the harvested area or from the surrounding forest. In the Grande Prairie region, the sites were clearcut harvested in winter 2003–2004 following the fall 2003 masting event; thus the preharvest spruce trees on and adjacent to the cutover were the primary seed source. In the Peace River region, the sites were clearcut harvested in the winter of 2005–2006 following a fall 2005 masting event. All surveys were conducted in 2006; thus, the Peace River sites were surveyed in the first growing season following harvest while the sites in the other regions were sampled in the third growing season following harvest. Varying numbers and lengths of transects were laid out in each region; the number of cutblocks, number of transects, the range of lengths for individual transects, the total length of all transects combined, and total number of quadrats assessed were as follows: Drayton Valley, 6 cutblocks, 15 transects, 50–310 m, 2000 m, 347 quadrats; Edson, 4 cutblocks, 5 transects, 105–230 m, 935 m, 109 quadrats; Grande Prairie, 8 cutblocks, 11 transects, 45 m, 495 m, 110 quadrats; and Peace River, 5 cutblocks, 14 transects, 45 m, 630 m, 110 quadrats. These deciduous cutblocks were approximately 10 ha in size. Sampling was conducted within 1 m² quadrats every ~5 m along each transect. For each white spruce seedling present in a quadrat, the substrate type that it was rooted in was recorded (as per Table 1). Substrate availability was determined by recording the substrate at four randomly selected locations around the edge of each quadrat. From these data, we calculated counts of seedling presence or absence for each microsite in each region.

Seedling substrate preferences were quantified by creation of categorical models of seedling presence (vs. absence) as a function of substrate and region. In initial runs, the substrate × region interaction was significant, so we subsequently ran models for each region separately. Estimates calculated from the following model were used to quantify seedling preference for each sub-

strate: model logit (seedling present) = intercept + $\beta_1 X_1 + \beta_2 X_2 + \dots$, where β_1 is the estimate for the influence of substrate 1, β_2 is the estimate for the influence of substrate 2, etc. These analyses were conducted using PROC CATMOD in SAS version 9.3 (SAS Institute, Inc. 2011).

Part III: surrounding vegetation effects on survival and growth

Each seedling encountered in the surveys in 2006, as described in Part II, was tagged, and its total height and leader height were recorded. These seedlings were re-visited in July 2007 for assessment of the surrounding vegetation and in September 2007 for determination of growth and survival. In Peace River, this was the second growing season after harvesting; at Edson, Drayton Valley, and Grande Prairie, this was the fourth growing season after harvesting. Subsequent analyses focused on survival from 2006 to 2007, and leader length in 2007 (for surviving seedlings) was used as the measure of growth. Fifteen different measures of surrounding vegetation were determined for each seedling as follows: the number of large saplings (>1.3 m height, mostly trembling aspen) and percent cover of intermediate saplings (<1.3 m but >1 m height) within a 10 m² circular plot centered on the 1 m² quadrat; percent cover of small saplings (<1 m height) in the 1 m² quadrat; percent cover of tall forbs (>30 cm height), short forbs (<30 cm height), live graminoids, and live and dead graminoids in a 30 cm × 30 cm quadrat centered on the seedling; and cover of all saplings, forbs, live graminoids, and live and dead graminoids in an imagined vertical cone of a 90° angle centered over the top of the seedling. Total live cover was the summed cover of all saplings, forbs (both heights), and live graminoids from the quadrats, while total live and dead cover was “total live cover” plus dead graminoid cover. Similarly, total live cover and total live and dead cover in the vertical cone were calculated in the same way.

For analyses, each seedling was considered a sampling unit, and quadrat within transect was treated as a random factor; each region was analyzed separately. For analyses of mortality, we used mixed-model ANOVA to compare the cover for each vegetation category between plots with living seedlings versus those in which seedlings died over the 1-year period of assessment. When there was a significant difference, we examined least-squares means (and standard errors) for plots with live versus dead seedlings to determine the direction of the difference. These analyses

were conducted using PROC MIXED in SAS version 9.2 (SAS Institute, Inc. 2011).

We developed models for the influence of surrounding vegetation on seedling growth (leader length in 2007) in each region as follows. First, mixed-model ANOVAs were run for each vegetation variable to determine the significance of the influence on seedling growth (2007 leader length); for these, the vegetation variable was the (continuous) fixed effect and transect was included as a random (block) effect. Model residuals were tested and square root transformations were applied only to Drayton Valley to obtain normalized residuals. Then we determined pairwise correlations between the vegetation variables. When variables were significantly correlated, we retained the one that had the strongest influence on the response variable (seedling growth) and discarded the other. The retained, uncorrelated (correlation coefficient < 0.20) variables were then used to build mixed models for seedling growth as a function of the different vegetation variables, with transect as a random term. These models were constructed using the “dredge” function (“MuMIn” package (Barton 2015) in R version 2.15.1; R Core Team 2015); the final best model was chosen based on Akaike Information Criterion (AIC). Residuals for final models were checked for evidence of nonlinearity, non-normality, or heteroscedasticity; it was confirmed that no further transformations were required.

For the Peace River site only, seedlings were visited in spring 2007 at the beginning of the second growing season, at which time we noted for each seedling whether it was covered by leaf litter (no cover, partially covered, completely covered) and whether it was submerged in water (yes or no) as a result of a large snowmelt. To determine the influence of submergence and cover by leaf litter on seedling mortality at the Peace River site, we used a categorical model in which the logit of seedling survival was modeled as a function of the influence of submergence, cover by litter, and their interaction. These analyses were conducted using PROC CATMOD in SAS version 9.2 (SAS Institute, Inc. 2011).

Results

Part I: substrate effects on early establishment of sown seeds

There were significant differences among the substrates in terms of early establishment success in the first summer following sowing of white spruce seed for both the deciduous- and conifer-leading forest sites (Table 2). On the ripper-ploughed conifer-dominated sites, the shelf position (composed of loosened mineral soil on the upper side of the scarified trench) and scraped rotten wood resulted in the highest early seedling establishment of white spruce (Table 2). Other substrates created by site preparation — scalped mineral soil, exposed organic layer (FH), and lower trench positions — had moderately successful early establishment (Table 2). There was essentially no natural regeneration on the mound and undisturbed moss and litter substrates. On both the scalped and mound microsites, we observed dead seedlings laying horizontal with exposed roots. On deciduous-leading clearcut sites with no site preparation, spruce seeds sown on rotten wood had significantly better early establishment than on any other substrate examined; success was negligible on moss, thin and thick organic layers, and exposed mineral soil (Table 2).

Part II: substrate effects on establishment of natural seedlings

In logged, deciduous-leading mixedwood sites in all four regions, the most common of the naturally available surface substrates was thick organic material (>5 cm depth) (Fig. 1a). Solid wood, rotten wood, and thin organics were each found to cover less than 20% of the surveyed sites in all regions, while the least available substrates in these deciduous stands were postdisturbance mosses and exposed mineral soil (Fig. 1a).

Table 2. Establishment of seeded white spruce (survival to the end of the first growing season) sown on different substrates (see Table 1).

	Survival (% of seeds sown)			
	Mean	Median	Range	Significance
Conifer				
Mound	0.08	0.00	0–1.67	d
Rotten wood	10.00	5.00	0–60	ab
Moss	0.04	0.00	0–1.67	d
Litter	0.00	0.00	0–0	d
Shelf	10.54	5.83	0–45	ab
FH	5.08	1.67	0–30	bc
Scalped	4.00	1.67	0–23.3	c
Trench	2.46	0.00	0–26.7	cd
Deciduous				
Rotten wood	1.54	0.00	0–8.33	a
Moss	0.00	0.00	0–0	b
Thick organic	0.07	0.00	0–1.67	b
Thin organic	0.13	0.00	0–1.67	b
Mineral soil	0.11	0.00	0–1.67	b

Note: Given is the mean, median, and range as a percentage of seeds sown for harvested conifer-leading sites with mechanical site preparation (Conifer) and harvested broadleaf-leading sites without site preparation (Deciduous). In harvested conifer-leading sites, litter and moss (feather moss) were naturally occurring substrates while the others were the result of site preparation. Lowercase letters denote significant differences ($p \leq 0.05$) between substrates within a forest site type.

The occurrence of natural white spruce regeneration in these four regions mirrored substrate availability, with the exception of solid wood on which there was almost no natural regeneration (Figs. 1a and 1b). Most seedlings were found on thick (>5 cm) organic surface material, which was the most available substrate, followed by thin organics (<5 cm deep) and then mineral soil and postdisturbance mosses (Fig. 1b). Only in the Grande Prairie region was there substantial natural regeneration on rotten wood (Figs. 1a and 1b).

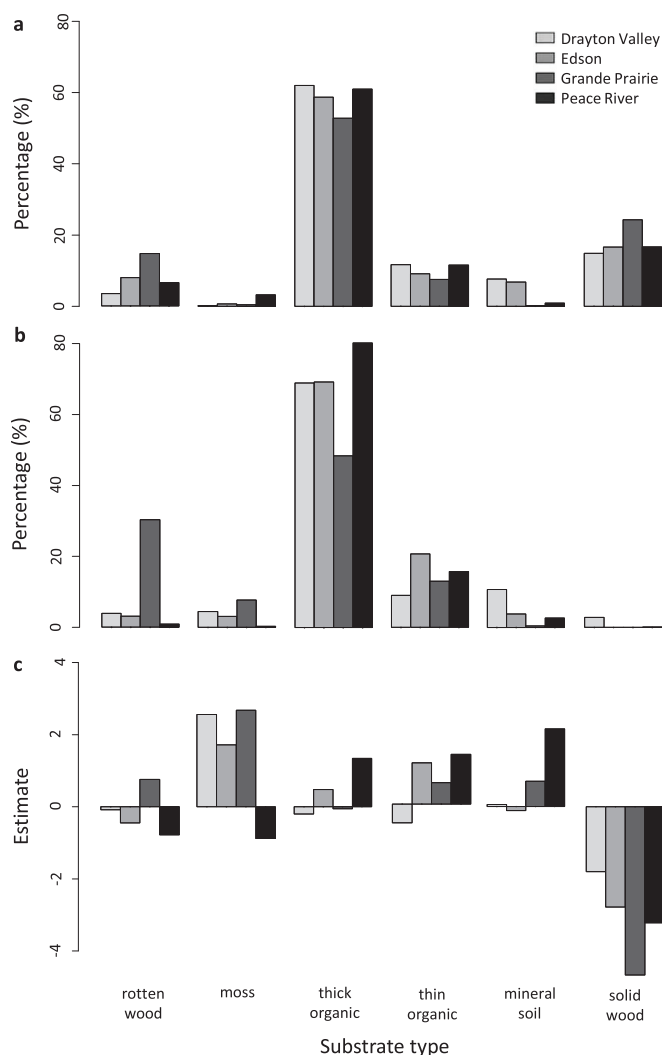
Substrate preference indicates how often a substrate was occupied by a spruce seedling relative to its availability (results from categorical models) (Fig. 1c). These results show that postdisturbance mosses (e.g., *Polytrichum* spp.) was the most strongly preferred substrate for white spruce seedling establishment in all regions except Peace River, where surveys were conducted in the first postharvest growing season (Fig. 1c). In Peace River, exposed mineral soil was the most strongly preferred substrate. Edson was the only region where mineral soil was not a preferred substrate for seedling establishment, although the results show only very weak avoidance (Fig. 1c). The thin organic substrate was preferred in all regions except Drayton Valley, where it was slightly avoided (Fig. 1c). Effects of the thick organic and rotten wood substrates on the establishment of white spruce seedlings were highly variable by region (Fig. 1c). Solid wood was always strongly avoided (Fig. 1c).

Part III: surrounding vegetation effects on survival and growth

Mortality

Comparisons of the effects of competing vegetation on seedling mortality indicated that effects of vegetation differed among regions (Table 3). In Grande Prairie, forb abundance (cover in cone and short forb cover) was higher in plots in which seedlings survived (from the third to the fourth postharvest growing season), while live graminoid cover was negatively associated with seedling survival in this time period. Similarly, in Edson, cover of short forbs was positively associated with survival of white spruce seedlings. In Drayton Valley, all four measures of graminoid cover, as well as three of the measures of total live and total live and dead cover, were significantly negatively associated with seedling survival (Table 3). Results were different in Peace River, where sur-

Fig. 1. Substrate availability and seedling preference based on surveys of natural regeneration of white spruce in unprepared deciduous-leading boreal mixedwood forests in the first (Peace River) or third (other sites) growing season after harvesting: (a) availability of six different substrate types as a percentage of all substrates, separately for each region; (b) percentage of white spruce seedling in the six substrate types, for each region; and (c) substrate preference for white spruce seedlings as indicated by the estimate (from a categorical model) for the influence of substrate on the probability of encountering a naturally established seedling. Positive values indicate preferred substrates; negative values indicate nonpreferred substrates. See Table 1 for descriptions of substrates.



vival was assessed from the first to the end of the second postharvest growing season. Here, all four measures of sapling (primarily aspen suckers) abundance were higher in plots in which seedlings survived (Table 3). In the Peace River site, an unusual event allowed us to assess the survival of young spruce seedlings after a temporary spring flooding event. This site also had dense aspen suckers resulting in a thick leaf litter layer from the previous fall. Both temporary spring flooding and especially cover by aspen leaf litter in the spring reduced seedling survival; cover by leaf litter reduced survival to half for seedlings that were not submerged and to one-quarter for submerged seedlings (Table 4).

Growth

In all regions except Edson, at least one vegetation variable had a significant effect on seedling growth (Table 5). Peace River had

the greatest number (nine) of significant vegetation variables for predicting growth (Table 5). Three of these were included in the final best model: number of saplings, forb cover in the cone, and total (live and dead combined) graminoid cover in the cone (Table 6). All three of these vegetation variables were positively related to seedling growth. In contrast, in the Grande Prairie and Drayton Valley regions, sapling abundance was negatively related to seedling growth (Table 5). For Grande Prairie, there was only one significant variable; sapling cover in the cone was negatively related to growth (Table 5). Live graminoid cover also had a negative influence and was tested for inclusion in the model; however, the best-fit model included only the negative influence of sapling cover in the cone (Table 6). Drayton Valley also only had one significant variable: cover of tall saplings was negatively related to growth (Table 5). Total vegetation in the cone was also tested for inclusion in the model. The best-fit model only included an intercept, so the second-best model — based on AIC — was used; it included the negative influence of cover of tall saplings (Table 6). For Edson, none of the vegetation variables was significant, so no model was constructed (Table 5).

Discussion

What substrates best support white spruce establishment?

Our sowing experiment showed that dead and dying feather mosses were poor substrates for establishment. Feather mosses are common on conifer sites but rare in deciduous forests (Startsev et al. 2008). The unsuitability of feather mosses as an establishment substrate is likely related to their coarse structure. They act as a thick physical barrier preventing seedling roots from reaching the mineral soil (Nienstaedt and Zasada 1990; Nilsson et al. 1996; Gärtner et al. 2011), and they could also inhibit establishment through chemical interference (Nilsson et al. 1996). Feather mosses die out in the years following disturbance (Nilsson et al. 1996; Wurtz and Zasada 2001) and are replaced by thinner weedy mosses such as *Polytrichum* spp., which can also produce a dense cover on exposed mineral soil. These were a preferred substrate for establishment in the first 3–4 years in all deciduous sites except the 1- to 2-year-old Peace River site, where these mosses were not abundant. These weedy, postdisturbance mosses may act as phytometers, occupying the sites where mineral soil stays consistently moist and growing conditions are reliable. These mosses are more dense and shorter in stature than feather mosses, likely allowing the seeds and early germinants to be in closer contact with a more stable moisture source in the mineral soil while accessing sufficient light (DeLong et al. 1997; Wang and Kembell 2005). Similarly, thin (as compared with thick) organic substrates can hold surface moisture and provide nutrients while allowing seedling roots easier access to the mineral soil layer.

If access to mineral soil is so important for seedling establishment, our results that exposed mineral soil was not a preferred substrate for natural regeneration would seem contradictory. A synthesis of 30 records from 11 studies of white spruce early seedling survival on mineral soil substrates showed in a mean survivorship of 0.082 (Greene and Johnson 1998), which is in line with our value of 0.11 for deciduous sites. Greene and Johnson (1998) noted, however, the strong right skew in the data (which we also observed) and the exceptionally high variation among studies, which they suggested might reflect interannual variation in drought. There are a number of other reasons why pure mineral soil substrates might sometimes be unsuitable for seedling establishment, including low nutrient availability, susceptibility to erosion, high temperatures, and propensity for frost heaving (Norberg et al. 2001; Mallik and Kravchenko 2016). These could explain our observed poor suitability of mineral soil substrates in the surveys of natural regeneration and in the sowing experiment, as well as the poor suitability of loose mineral soil substrates created by mounding and trenching of the conifer-leading

Table 3. Results of mixed-model ANOVAs comparing competing vegetation between plots with naturally regenerated white spruce seedlings (live versus dead) in postharvest deciduous-leading boreal mixedwood forests in four regions.

	Peace River (age 1–2) ^a		Grande Prairie (age 3–4) ^a		Drayton Valley (age 3–4) ^a		Edson (age 3–4) ^a	
	<i>p</i>	Effect	<i>p</i>	Effect	<i>p</i>	Effect	<i>p</i>	Effect
No. of saplings	<0.0001	+	0.2287		0.5490		0.1819	
Tall sapling cover	0.0010	+	0.0777		0.9998		0.1982	
Sapling cover in cone*	0.0028	+	0.4250		0.6525		0.4071	
Short sapling cover	0.0147	+	0.6361		0.2505		0.5741	
Tall forb cover	0.0656		0.0770		0.5128		0.6452	
Forb cover in cone*	0.4756		0.0170	+	0.5525		0.3124	
Short forb cover	0.2641		0.0027	+	0.8960		0.0262	+
Live graminoid cover	0.2004		0.0431	–	0.0001	–	0.3257	
Live graminoid cover in cone*	0.3907		0.1774		<0.0001	–	0.3631	
Live and dead graminoid cover	0.0968		0.0918		0.0004	–	0.1569	
Live and dead graminoids in cone*	0.3100		0.0541		<0.0001	–	0.2608	
Total live cover	0.7958		0.1133		0.3798		0.5475	
Total live cover in cone*	0.5463		0.0841		0.0437	–	0.2414	
Total live and dead cover	0.9456		0.5799		0.0225	–	0.5091	
Total live and dead cover in cone*	0.7910		0.0841		0.0016	–	0.8614	
No. of seedlings sampled	878		114		465		136	

Note: Vegetation was assessed in plots centered on the seedling. An asterisk (*) indicates cover that was estimated in an imagined vertical cone of a 90° angle centered over the top of the seedling (see Methods for further details). Significant variables in each model ($p \leq 0.05$) are indicated in bold. Effect: “+” indicates a variable that was positively associated with seedling survival; “–” indicates a variable that was negatively associated with survival.

^aThe time period in years after harvest over which survival was assessed.

Table 4. Results showing the influence of cover by leaf litter and submergence in water on the probability of seedling survival at the deciduous-leading mixedwood site at the Peace River site.

Proportions of seedlings surviving in different submergence and litter cover conditions from the survey of 865 seedlings at the Peace River site in spring 2007

Leaf cover	Submerged	No. of seedlings	Proportion surviving
No	No	334	0.7575
Partial	No	169	0.7811
Yes	No	181	0.3536
No	Yes	52	0.7115
Partial	Yes	21	0.4762
Yes	Yes	108	0.1852

Results of categorical models showing the significance of the influence of submergence and cover by leaves on survival

Source	df	χ^2	<i>p</i>
Intercept	1	3.05	0.0807
Covered	2	94.85	<0.0001
Submerged	1	14.63	0.0001
Covered × submerged	2	4.28	0.1179

Note: df, degrees of freedom.

sites at Peace River. Logging tends to expose mineral soil along skid trails (Solarik et al. 2010) where heavy machinery and tree removal scrape the forest floor, churn up duff layers, and compact soils (Wurtz and Zasada 2001; Martin-DeMoor et al. 2010; Gärtner et al. 2011). While scraped mineral soil substrates may be suitable for seedling establishment, churned layers would, like the cast-off soil from site preparation, be loose and prone to desiccation, washing-out, and sometimes frost-heaving. On the other hand, compacted soils would have poor root penetration in addition to sharing characteristics with prepared trenches, which are prone to cold-air pooling and flooding. It seems that the suitability of mineral soil and other substrates for *P. glauca* establishment is highly context-dependent.

Submergence combined with a covering of aspen leaf litter appeared to be particularly lethal to many spruce seedlings (80%

seedling mortality), while on its own, flooding had little effect. Flooding can prevent seedling roots from accessing oxygen, causing seedling mortality (Landhäusser et al. 2003), but our results show that small white spruce seedlings can survive short periods of submergence. Their survival seems to be more strongly negatively affected by the heavy litterfall from the dense aspen suckers that establish on these sites. These shed a thick layer of leaves that cover small spruce seedlings, blocking sunlight, smothering them, and increasing mortality from snow press (heavy snow loading causing the grass to collapse) (DeLong et al. 1997). Our results show that partial cover by litter does not hamper survival but complete coverage dramatically reduces survival; indeed, over half of the overall mortality of seedlings at Peace River was attributable to cover by litter (71% mortality for seedlings with heavy litter cover, with or without submergence, as compared with overall mortality of 40%; Table 4). The influence of cover by litter was even more dramatic when the seedlings were already experiencing the stress of submergence. Litterfall is likely a reason for the limited period of recruitment of spruce seedlings immediately after disturbance in some stands (Peters et al. 2006); there is a small window of opportunity for these very small seedlings to establish before aspen suckers begin shedding thick layers of leaf litter.

As expected, rotten wood was found to be one of the best substrates for first-year success of seeded white spruce; however, it was not preferred for natural seedling establishment on harvested deciduous-dominated sites. Overall, increased exposure in harvested areas combined with logging traffic that tends to break up and scatter rotten wood pieces (Wang and Kemball 2005; Johnstone and Chapin 2006) likely makes rotten wood a less suitable substrate in these conditions as smaller pieces may become exposed and be more susceptible to desiccation than intact rotten logs. Rotten wood that is capable of supporting seedling establishment may therefore be a small percentage of all rotten wood available on a postlogged site; hence, it likely plays a minor role in harvested areas compared with what would be expected on undisturbed sites.

Table 5. Results of mixed linear models examining the effects of competing vegetation on growth (leader length in 2007) of naturally regenerated white spruce in deciduous-leading boreal mixedwood forests in four regions.

	Peace River (2nd season) ^a		Drayton Valley (4th season) ^a		Grande Prairie (4th season) ^a		Edson (4th season) ^a	
	<i>p</i>	Effect	<i>p</i>	Effect	<i>p</i>	Effect	<i>p</i>	Effect
No. of saplings	0.0020	+	0.0671		0.0682		0.1650	
Tall sapling cover	0.0542		0.0332	–	0.0861		0.2641	
Sapling cover in cone*	0.4884		0.0713		0.0033	–	0.5228	
Short sapling cover	0.7729		0.7211		0.0798		0.4691	
Tall forb cover	0.1415		0.1601		0.4942		0.6714	
Forb cover in cone*	0.0145	+	0.5458		0.5301		0.8954	
Short forb cover	0.0229	+	0.7458		0.8627		0.9239	
Live graminoid cover	0.0466	+	0.2531		0.0577		0.8688	
Live graminoid cover in cone*	0.0009	+	0.4084		0.2286		0.4595	
Live and dead graminoid cover	0.0009	+	0.1394		0.4907		0.4647	
Live and dead graminoid cover cone*	0.0001	+	0.1415		0.5112		0.5049	
Total live cover	0.0242	+	0.9584		0.5510		0.4150	
Total live cover in cone*	0.0192	+	0.7688		0.9498		0.6277	
Total live and dead cover	0.0529		0.8462		0.6843		0.6179	
Total live and dead cover in cone*	0.4329		0.0886		0.5790		0.5568	
No. of seedlings (live) sampled	562		98		331		91	

Note: Vegetation was assessed in plots centered on the seedling. An asterisk (*) indicates cover that was estimated in an imagined vertical cone of a 90° angle directly above the seedling (see Methods for further details). Significant variables in each model ($p \leq 0.05$) are indicated in bold. Effect: “+” indicates a variable that was positively associated with seedling survival; “–” indicates a variable that was negatively associated with growth. Drayton Valley growth measurements were square root transformed prior to analysis to meet assumptions of normality and homogeneity of variance of residuals.

^aThe postharvest growing season in which growth was assessed.

Table 6. Final models for seedling growth (*G*) in relation to measured vegetation variables.

Region	Sample size	Model	SE	<i>p</i>	<i>r</i> ²
Peace River	562	$G = 0.211\text{Sn} + 0.130\text{Fc} + 0.169\text{Gc} + 7.306$			0.2672
		Sn, number of saplings	0.0719	0.0036	
		Fc, forb cover in cone	0.0417	0.0020	
		Gc, all graminoids in cone	0.0404	0.0000	
		Intercept	1.9456	0.0002	
Grande Prairie	98	$G = -0.854\text{Sc} + 49.900$			0.3898
		Sc, saplings in cone	0.2781	0.0033	
		Intercept	5.9574	0.0000	
Drayton Valley	331	$G = -0.01788\text{St} + 4.88$			0.4724
		St, cover of tall saplings	0.0083	0.0332	
		Intercept	0.2693	0.0000	

Note: The model and its R^2 values are given along with the standard error for the estimate (SE) and significance (*p*) for each variable in the model. No model was constructed for the Edson region because none of the vegetation variables was significant (see also Table 5).

How does competing vegetation affect the growth and mortality of natural white spruce regeneration?

Surrounding vegetation was not always negatively associated with seedling survival and growth for these very young seedlings, a result supported by some previous studies (e.g., Heineman et al. 2005; Man et al. 2008), but not all. Heineman et al. (2005) found a similar result where planted interior spruce (*Picea glauca* × *englemanni*) on mesic sites did not grow better with vegetation control during the 5 years studied. We found that survival was positively associated with abundance of saplings (mostly aspen suckers) or forb cover in three of the four regions. Similarly, growth of the very youngest seedlings (in Peace River) was positively associated with neighboring vegetation cover, including saplings, forbs, and graminoids. Neighbouring vegetation, however, is not universally beneficial for seedling survival and growth (Landhäusser et al. 1996; Man et al. 2008), and there was a tendency for it to have a negative influence in Drayton Valley, where the vegetation cover was greatest. The positive association between abundance of surrounding vegetation and seedling survival or growth could simply reflect the fact that both were responding to site conditions. Within the median range of growing conditions, good sites would have

higher cover of neighbouring vegetation as well as higher rates of white spruce survival and growth. Similarly, poor sites would support neither seedlings nor associated vegetation; thus, lower seedling survival and growth would be observed together with lower vegetation cover.

There were also negative effects of surrounding vegetation on seedlings. Graminoid cover was negatively associated with seedling survival in Grande Prairie and Drayton Valley. Similar results have been found in other studies as well (Cater and Chapin 2000; Man et al. 2008), and this suggests that a mechanism other than growing site may be influencing mortality. While small seedlings may be able to survive quite well under grass, the influence of snowfall may have a significant effect in this situation. Snow press can crush the seedling below the snow, causing growth deformities and mortality (Eis 1981). Also, thick grass layers can prevent the ground from warming quickly in the spring, reducing the growing season for small seedlings (Cater and Chapin 2000; Man et al. 2008; Pitt et al. 2010). Further, dense grass cover can harbour large rodent populations (Cater and Chapin 2000), including voles, which are known to cause seedling mortality by consuming bark and girdling seedlings. Higher cover of broadleaf saplings

may reduce this grass cover and increase exposure, thereby benefiting spruce seedling survival (Man et al. 2008; Pitt et al. 2010); however, this would only be for seedlings that are tall enough to withstand the press effect of leaf litter.

Conclusions

Feather mosses, thick organic layers, litter, and solid wood were poor substrates for establishment of white spruce seedlings in postharvest boreal mixedwood forests. Thin organic layers and early successional mosses establishing on mineral soils facilitated white spruce regeneration. This is likely attributable to these substrates providing correct moisture conditions. Where these substrates are found in abundance, site preparation may be unnecessary to aid in reforestation. Sites with thick organic layers and feather mosses should be considered for site preparation treatments to increase the quantity of preferred substrates if natural regeneration of white spruce is desired. Rotten wood was not a consistently preferred substrate; its suitability is likely negatively impacted by logging operations that breakup deadwood, exposing it to desiccation. It thus should not be relied on to support natural regeneration in the immediate postharvest period. However, we know that in the longer term, it can act as an excellent substrate to support ongoing establishment under a developing forest canopy; thus leaving snags and other standing trees for future recruitment of deadwood is advisable. In several cases, surrounding vegetation was positively associated with survival and growth, especially of the youngest seedlings. This suggests that in the early postharvest period, spruce seedlings might benefit from shelter, indicating that moderate levels of associated vegetation are good for establishment; however, on older sites or where competition is severe, competing vegetation and especially heavy litterfall from aspen saplings is likely to reduce both survival and growth of small white spruce seedlings. Our results demonstrate that white spruce is able to successfully regenerate naturally in a range of conditions that exist on postharvest boreal mixedwood sites, beyond those created by fire or that exist under an established canopy as a part of secondary succession. With greater insight, the use of natural regeneration to establish white spruce is a potentially successful management strategy for boreal mixedwood forest regeneration following harvesting.

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